Automotive Technical Service Diagnostic Bulletin

Document ID: DIAG-S-CLASS-P0301
Classification: OEM Level 3 Diagnostic & Repair Procedure
Target Vehicle: Mercedes-Benz S-Class (W223 / S 500 4MATIC)
Powertrain: 3.0L Turbocharged Inline-6 (M256) with 48V Integrated Starter-Generator (ISG) & Electric Auxiliary Compressor (eZV)
Engine Control System: Bosch MG1CP002 (Engine Control Module N3/10) - FlexRay / CAN-FD Gateway
Electrical Architecture: Dual 12V / 48V Mild-Hybrid Architecture
Fuel Injection Architecture: 200-350 bar Direct Injection (Piezo HP Fuel Injectors, dual HP fuel pumps)
Firing Order: 1-5-3-6-2-4

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1. DIAGNOSTIC TROUBLE CODE (DTC) SPECIFICATION

- **DTC Code:** `P0301`
- **Standard Fault Description:** Cylinder 1 Misfire Detected
- **System Definition:** Combustion failure or poor flame propagation specifically localized to Cylinder 1 based on CKP sector timing delta.
- **MIL (Malfunction Indicator Lamp) Status:** Illuminated (Continuous or Flashing depending on catalyst severity)
- **Monitoring Condition:** Continuous monitoring during engine running cycle with coolant temp > 60°C and vehicle speed > 0 km/h.

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2. PLATFORM-SPECIFIC ROOT CAUSE ANALYSIS (RCA)

Root cause analysis customized specifically for the **Mercedes-Benz S-Class (W223 / S 500 4MATIC)**:

1. **Piezo Injector Crystal Degradation on Cylinder 1:** Piezo actuator stack leakage or internal valve needle sticking caused by thermal fatigue on M256 aluminum cylinder head.
2. **Individual Coil T1/1 Insulation Breakdown:** Corona discharge through the silicone boot extension down the deep M256 spark plug well.
3. **Intake Valve Carbon Build-up (GDI):** Excessive carbon deposit on cylinder 1 intake port reducing swirl coefficient at cold start.

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3. DEDICATED WIRING DIAGRAM & CIRCUIT PINOUT

Schematic configuration and electrical pin connections specifically for the S-Class control harness:

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[N3/10 ECM Connector 'M']                  [Piezo Injector Y76/1]
Pin 45 (Cyl 1 High-Side +65V/140V) ---> Pin 1 (Actuator +)
Pin 46 (Cyl 1 Low-Side Control)    ---> Pin 2 (Actuator -)

[N3/10 ECM Connector 'E']                  [Coil On Plug T1/1]
Pin 12 (Logic Trigger 5V) ----> Pin 3 (ECU Signal)
Circuit 87M (Protected via Fuse 104) -> Pin 1 (12V Switched B+)
Ground Point W10/4 (Chassis Head)     -> Pin 2 (Signal/Power GND)
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- Resistance Spec: Piezo Injector terminals = ~180kΩ - 220kΩ (Never connect direct 12V; high voltage piezo driver).

Electrical Reference Points

- **Primary Ground Reference:** `W10/4 (Cylinder Head Ground, Left) & W16/5 (Front Bulkhead Ground)`

- **Power Distribution Node:** `Front SAM (Signal Acquisition Module N10/6) Fuse Box F32/3`
- **Signal Logic Voltage Range:** 0.0V to 5.0V regulated DC (unless high-voltage direct injection rail)

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4. STEP-BY-STEP DIAGNOSTIC & VERIFICATION PROCEDURE

Follow this OEM workshop sequence to isolate the failure on this platform:

1. Swap Coil T1/1 to Cylinder 5 to see if DTC follows the coil.
2. Using an oscilloscope with high-voltage differential probe, capture Injector Y76/1 voltage profile (Target: 140V peak pulse).
3. Perform relative compression test via XENTRY starter-draw motor rpm diagnostic on Cylinder 1.
4. Bore-scope cylinder 1 to inspect combustion bowl and spark plug electrode gap (Spec: 0.70mm ± 0.05mm).

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5. PRECISE RESOLUTION & REPAIR PROCEDURE

- Replace faulty Piezo Injector Y76/1 using Mercedes special tool W256 589 01 33 00.
- Mandatory: Code new IMA/QR classification code (16-digit hexadecimal code) into N3/10 ECM adaptation table.
- Replace spark plug on Cylinder 1 (Torque strictly to 23 Nm; indexing direction towards injector).

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6. POST-REPAIR QUALITY VALIDATION & RE-TESTING

1. Clear all DTCs from Engine Control Module (`Bosch`) and secondary gateway modules.
2. Perform comprehensive drive cycle:
3. Cold start idle for 3 minutes (monitors secondary air / catalyst preheat and cold misfire).
4. Accelerate smoothly to 50 km/h and hold for 2 minutes.
5. Cruise at 80–100 km/h on highway for 10 minutes (validates downstream O2 sensor and fuel trim closed-loop balance).
6. Coasting deceleration (fuel cut-off check) for 5 seconds without braking.
7. Re-scan control units with diagnostic tool; confirm all OBD-II readiness monitors show **READY** / **COMPLETE** and no pending DTCs are stored.